10.4.7 Differentiation of one function with respect to another function :

If $y = f(x)$ is differentiable, then the derivative of y with respect to x is

$$\frac{dy}{dx} = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} .$$

If f and g are differentiable functions of x and if $\frac{dg}{dx} = g'(x) \neq 0$, then

$$\frac{df}{dg} = \frac{\frac{df}{dx}}{\frac{dg}{dx}} = \frac{f'(x)}{g'(x)}.$$

Example 10.28

Find the derivative of x^x with respect to $x \log x$.

Solution

Take $u = x^x, v = x \log x$

$$\log u = x \log x$$

$$\frac{1}{u}\frac{du}{dx} = x.\frac{1}{x} + 1.\log x = 1 + \log x$$

$$\frac{du}{dx} = u(1 + \log x) = x^x(1 + \log x)$$

$$\frac{dv}{dx} = 1 + \log x$$

$$\frac{d(x^x)}{d(x \log x)} = \frac{du}{dv} = \frac{\frac{du}{dx}}{\frac{dv}{dx}} = x^x .$$

Note that when g is the identity function $g(x) = x$ then $\frac{df}{dg}$ reduces to $\frac{df}{dx} = f'(x)$.

Example 10.29

Find the derivative of $\tan^{-1}(1 + x^2)$ with respect to $x^2 + x + 1$.

Solution :

Let $f(x) = \tan^{-1}(1 + x^2)$

$$g(x) = x^2 + x + 1$$

$$\frac{df}{dg} = \frac{f'(x)}{g'(x)}$$

$$f'(x) = \frac{2x}{(x^4 + 2x^2 + 2)}$$

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